KOREATECH
한국기술교육대학교

Variational Perspective: From VAEs to DDPMs – I

March 2026

https://linklab.github.io

Variational Autoencoder

Basic Concepts of Variational Autoencoders

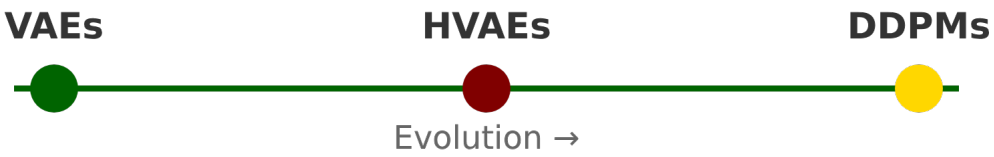
Limitations of Standard Autoencoders

- Deterministic encoding produces **unstructured latent space** (random latent codes)
- Random latent codes typically produce **meaningless outputs**
- Limited use for generation tasks

VAE's Probabilistic Approach

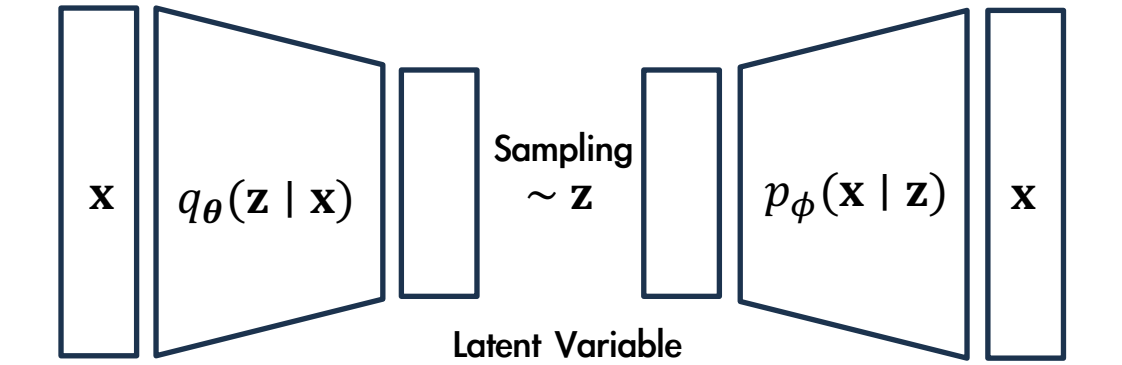
- → Imposes **probabilistic structure** on latent space
- → Stochastic encoder $q_\theta(\mathbf{z} \mid \mathbf{x})$ and decoder $p_\phi(\mathbf{x} \mid \mathbf{z})$
- → Latent space becomes **smooth and continuous**
- → Foundation for Advanced Models

VAEs: **true generative models**



VAE framework extends to **hierarchical models** (HVAEs) and **diffusion models (DDPMs)**

Probabilistic Encoder and Decoder in VAEs



Prior Distribution

$$\mathbf{z} \sim p_{prior} = \mathcal{N}(0, 1)$$

- Latent variables follow standard Gaussian prior

Decoder Distribution

$$p_\phi(\mathbf{x} \mid \mathbf{z})$$

- Maps latent variables to data space

[Key Relationships in VAEs]

Probabilistic Encoding
$q_\theta(\mathbf{z} \mid \mathbf{x})$ maps data to latent

Probabilistic Decoding
$p_\phi(\mathbf{x} \mid \mathbf{z})$ generates data from latent

Latent–Data Relationship
Connects observed and latent variables

VAE Decoder Construction and Marginal Likelihood

Latent-Variable Generative Model

The VAE defines a latent-variable generative model through the marginal likelihood:

$$p_\phi(\mathbf{x}) = \int p_\phi(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z})\ d\mathbf{z}$$

x가 나올 확률은, 모든 가능한 z가 x를 만들 확률을 합친 것

- Observed data $\mathbf{x}$ generated from latent variable $\mathbf{z}$
- Latent prior $p(\mathbf{z})$ typically standard Gaussian

Ideally $\max_\phi log p_\phi(\mathbf{x})$

Computational Challenge

- The decoder parameters ϕ at $p_\phi(\mathbf{x} \mid \mathbf{z})$ are learned by MLE
- The integral over $\mathbf{z}$ is intractable
- Direct MLE at $p_\phi(\mathbf{x})$ is computationally infeasible

Variational Approach

[Reverse question] given an observation $\mathbf{x}$, what latent codes $\mathbf{z}$ could have produced it?

Posterior

$$p_\phi(\mathbf{z} \mid \mathbf{x}) = \frac{p_\phi(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z})}{p_\phi(\mathbf{x})}$$

데이터 x에 대해, 우리가 궁금한 것 → 이 x는 대체 어떤 z에서 생성될 것일까? → Posterior Distribution → 학습할 때는 Posterior를 통하여 x를 잘 설명하는 z가 무엇인지 알아야 decoder가 올바로 학습됨

- By Bayes' rule, the posterior distribution over latent variables $\mathbf{z}$ given observation $\mathbf{x}$
- The denominator $p_\phi(\mathbf{x})$ still requires integrating over all latent variables

To address the computational infeasibility, we introduce an encoder (or inference network) $q_\theta(\mathbf{z} \mid \mathbf{x})$, whose role is to serve as a learnable approx. posterior and replace the intractable posterior:

$$q_\theta(\mathbf{z} \mid \mathbf{x}) \approx p_\phi(\mathbf{z} \mid \mathbf{x})$$

In practice, $q_\theta(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mu_\theta(x), \sigma_\theta(x))$

Evidence Lower Bound (ELBO) Derivation

Theorem 2.1.1: Evidence Lower Bound (ELBO)

For any data point $\mathbf{x}$, the log-likelihood satisfies:

$$\log p_{\phi}(\mathbf{x}) \geq \mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\phi}, \mathbf{x})$$

where the ELBO is given by:

$$\mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}) = \underbrace{\mathbb{E}_{\mathbf{z}\sim q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\phi}}(\mathbf{x} \mid \mathbf{z})\right]}_{\text{Reconstruction Term}} - \underbrace{\mathcal{D}_{\text{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p_{prior}(\mathbf{z})\right)}_{\text{Latent Regularization}} \quad \text{[2.1.1]}$$

[Proof]

$$\log p_{\phi}(\mathbf{x}) = \log \int p_{\phi}(\mathbf{x}, \mathbf{z})\, d\mathbf{z} = \log \int q_{\theta}(\mathbf{z} \mid \mathbf{x}) \frac{p_{\phi}(\mathbf{x}, \mathbf{z})}{q_{\theta}(\mathbf{z} \mid \mathbf{x})} d\mathbf{z}$$

$$= \log \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[\frac{p_{\phi}(\mathbf{x}, \mathbf{z})}{q_{\theta}(\mathbf{z} \mid \mathbf{x})}\right] \geq \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\phi}(\mathbf{x}, \mathbf{z})}{q_{\theta}(\mathbf{z} \mid \mathbf{x})}\right] = \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{z}) p_{\phi}(\mathbf{x} \mid \mathbf{z})}{q_{\theta}(\mathbf{z} \mid \mathbf{x})}\right]$$

$$= \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[p_{\phi}(\mathbf{x} \mid \mathbf{z})\right] + \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[\log p(\mathbf{z}) - \log q_{\theta}(\mathbf{z} \mid \mathbf{x})\right]$$

$$= \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[p_{\phi}(\mathbf{x} \mid \mathbf{z})\right] - \mathbb{E}_{\mathbf{z}\sim q_{\theta}(\mathbf{z}|\mathbf{x})}\left[\log q_{\theta}(\mathbf{z} \mid \mathbf{x}) - \log p(\mathbf{z})\right]$$

$$= \mathbb{E}_{\mathbf{z}\sim q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\phi}}(\mathbf{x} \mid \mathbf{z})\right] - \mathcal{D}_{\text{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p(\mathbf{z}))$$

Evidence Lower Bound (ELBO) Derivation

Theorem 2.1.1: Evidence Lower Bound (ELBO)

For any data point $\mathbf{x}$, the log-likelihood satisfies:

$$\log p_{\phi}(\mathbf{x}) \geq \mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\phi}, \mathbf{x})$$

where the ELBO is given by:

$$\mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}) = \mathbb{E}_{\mathbf{z}\sim q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\phi}}(\mathbf{x} \mid \mathbf{z})\right] - \mathcal{D}_{\text{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p_{prior}(\mathbf{z})\right) \quad [2.1.1]$$

accurate reconstruction of the data

regularizing the latent variables by keeping them close to a simple prior distribution $p_{prior}(\mathbf{z})$ (often Gaussian)

- 각각의 z로부터 x를 최대한 정확히 복원
- 각 데이터 x마다 서로 다른 z 산출 노력

z가 가능한 정보를 많이 담도록 노력

Trade-off

- 각 데이터 x 마다 산출되는 z의 정보는 $\mathcal{N}(0,1)$ 분포를 지니도록 제한

z가 가능한 정보를 압축하여 적게 담도록 노력

Key Insight

ELBO provides a tractable lower bound on log-likelihood, enabling optimization via gradients.

항목	Reconstruction	Latent KL
z 역할	정보 저장	분포 정규화
x 간 구분	강하게	약하게
분산	작게	크게
평균	자유롭게	0 근처

Information-Theoretic View: ELBO as a Divergence Bound

True modeling error: Original MLE Problem

Minimizing $\mathcal{D}_{\mathrm{KL}}\left(p_{\mathrm{data}}(\mathbf{x}) \| p_{\phi}(\mathbf{x})\right)$

Two Joint Distributions

- Generative joint: $p_{\phi}(x, z) = p(z)p_{\phi}(x \mid z)$
- Inference joint: $q_{\theta}(x, z) = p_{\mathrm{data}}(x)q_{\theta}(z \mid x)$

Comparing these distributions yields

$$\mathcal{D}_{KL}\left(p_{data}(x) \| p_{\phi}(x)\right) \leq \mathcal{D}_{KL}\left(q_{\theta}(x, z) \| p_{\phi}(x, z)\right) \quad \text{(2.1.2)}$$

sometimes referred to as the chain rule for KL divergence

Also, note that

$$\log p_{\phi}(\mathbf{x}) - \mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}) = \mathcal{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p_{\boldsymbol{\phi}}(\mathbf{z} \mid \mathbf{x})\right)$$

Joint KL divergence expands as:

$$\mathcal{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) \| p_{\boldsymbol{\phi}}(\mathbf{x}, \mathbf{z})\right)$$
$$= \mathcal{D}_{\mathrm{KL}}\left(p_{\mathrm{data}}(\mathbf{x}) \| p_{\boldsymbol{\phi}}(\mathbf{x})\right) + \mathbb{E}_{p_{\mathrm{data}}(\mathbf{x})}\left[\mathcal{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p_{\boldsymbol{\phi}}(\mathbf{z} \mid \mathbf{x})\right)\right]$$

- True modeling error: $\mathcal{D}_{\mathrm{KL}}\left(p_{\mathrm{data}}(x) \| p_{\phi}(x)\right)$
- Inference error: $\mathbb{E}_{p_{\mathrm{data}}(x)}\left[\mathcal{D}_{\mathrm{KL}}\left(q_{\theta}(z \mid x) \| p_{\phi}(z \mid x)\right)\right]$

Key Insight

Maximizing the ELBO corresponds to reducing inference error, ensuring training minimizes a meaningful part of the overall bound.

Gaussian VAE Implementation

Encoder

Conditional distribution:

$$q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \coloneqq \mathcal{N}\left(\mathbf{z}; \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}), \operatorname{diag}\left(\boldsymbol{\sigma}^2_{\boldsymbol{\theta}}(\mathbf{x})\right)\right)$$

- Maps observed data $\mathbf{x}$ to latent code $\mathbf{z}$
- Gaussian distribution with learned parameters
- Encoder parameters:
  $\boldsymbol{\mu}_{\boldsymbol{\theta}}: \mathbb{R}^D \to \mathbb{R}^d$ and $\boldsymbol{\sigma}_{\boldsymbol{\theta}}: \mathbb{R}^D \to \mathbb{R}^d_+$

Decoder

Conditional distribution:

$$p_{\boldsymbol{\phi}}(\mathbf{x} \mid \mathbf{z}) := \mathcal{N}\left(\mathbf{x}; \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{z}), \sigma^2 \mathbf{I}\right)$$

- Maps latent code $\mathbf{z}$ back to data $\mathbf{x}$
- Fixed small variance σ
- Decoder parameters: $\boldsymbol{\mu}_{\boldsymbol{\phi}}: \mathbb{R}^d \to \mathbb{R}^D$
- **Output**: Reconstruction $\mathbf{x} = \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{z})$, where $\mathbf{z} \sim q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x})$

Under this assumption,
the reconstruction term in the ELBO simplifies as

$$\mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\phi}}(\mathbf{x} \mid \mathbf{z})\right] = -\frac{1}{2\sigma^2}\mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}\left[\left\|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{z})\right\|^2\right] + C$$

➔ C is a constant independent of $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$

The ELBO objective

$$\min_{\boldsymbol{\theta},\boldsymbol{\phi}} \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}\left[\frac{1}{2\sigma^2}\left\|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{z})\right\|^2\right] + \underbrace{\mathcal{D}_{\mathrm{KL}}\left(q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \| p_{prior}(\mathbf{z})\right)}_{\text{a closed-form solution}},$$

VAE's Blurry Generation Problem

VAE Problem Reformulated

Consider a fixed Gaussian encoder and a decoder

$$q_{\theta}(\mathbf{z} \mid \mathbf{x}) := \mathcal{N}\left(\mathbf{z}; \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}), \operatorname{diag}\left(\boldsymbol{\sigma}_{\boldsymbol{\theta}}^2(\mathbf{x})\right)\right)$$

$$p_{\phi}(\mathbf{x} \mid \mathbf{z}) := \mathcal{N}\left(\mathbf{x}; \boldsymbol{\mu}_{\phi}(\mathbf{z}), \sigma^2 \mathbf{I}\right)$$

Optimizing the ELBO reduces to minimizing the expected reconstruction error

$$arg \min_{\boldsymbol{\mu}} \mathbb{E}_{p_{\text{data}}(\mathbf{x}) q_{\text{enc}}(\mathbf{z}|\mathbf{x})} \left[\left\| \mathbf{x} - \boldsymbol{\mu}(\mathbf{z}) \right\|^2 \right]$$

데이터 x를 encoder로 z를 만든 다음, decoder $\mu(z)$ 로 복원했을 때 x와의 오차를 최소화하는 decoder μ를 찾아라.

[Solution]

This is a standard least squares problem in $\mu(z)$, and its solution is given in closed form

$$\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\text{enc}}(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$$

$q_{\text{enc}}(\mathbf{x} \mid \mathbf{z})$ is the encoder-induced posterior on inputs given latents

$$q_{\text{enc}}(\mathbf{x} \mid \mathbf{z}) = \frac{q_{\text{enc}}(\mathbf{z} \mid \mathbf{x}) p_{\text{data}}(\mathbf{x})}{p_{\text{prior}}(\mathbf{z})}$$

$$\boldsymbol{\mu}^*(\mathbf{z}) = \frac{\mathbb{E}_{p_{data}(\mathbf{x})}[q_{\text{enc}}(\mathbf{z} \mid \mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{data}(\mathbf{x})}[q_{\text{enc}}(\mathbf{z} \mid \mathbf{x})]}$$

VAE's Blurry Generation Problem

[증명] $\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\text{enc}}(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$

$$= \int \mathbf{x} \cdot q_{\text{enc}}(\mathbf{x}|\mathbf{z})\, d\mathbf{x}$$

$$= \int \mathbf{x} \cdot \frac{q_{\text{enc}}(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x})}{p_{\text{prior}}(\mathbf{z})}\, d\mathbf{x}$$

$$= \frac{1}{p_{\text{prior}}(\mathbf{z})} \int \mathbf{x} \cdot q_{\text{enc}}(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x})\, d\mathbf{x}$$

$$= \frac{\int q_{\text{enc}}(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) \cdot \mathbf{x}\, d\mathbf{x}}{\int q_{\text{enc}}(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x})\, d\mathbf{x}}$$

$$= \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\text{enc}}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\text{enc}}(\mathbf{z}|\mathbf{x})]}$$

[구체적인 예시]

Dataset: MNIST 이미지 3장

- $\mathbf{x}_1$: "1", $\mathbf{x}_2$: "7", $\mathbf{x}_3$: "9"

Latent point: $\mathbf{z}^* = [0.5, 0.3]$

Encoder 확률:

- $q_{\text{enc}}(\mathbf{z}^*|\mathbf{x}_1) = 0.1$
- $q_{\text{enc}}(\mathbf{z}^*|\mathbf{x}_2) = 0.3$
- $q_{\text{enc}}(\mathbf{z}^*|\mathbf{x}_3) = 0.05$

최적 **Decoder** 계산

$$\boldsymbol{\mu}^*(\mathbf{z}^*) = \frac{0.1 \cdot \mathbf{x}_1 + 0.3 \cdot \mathbf{x}_2 + 0.05 \cdot \mathbf{x}_3}{0.1 + 0.3 + 0.05} = 0.22 \cdot \mathbf{x}_1 + 0.67 \cdot \mathbf{x}_2 + 0.11 \cdot \mathbf{x}_3$$

결과: "1" (22%) + "7" (67%) + "9" (11%) = 블러

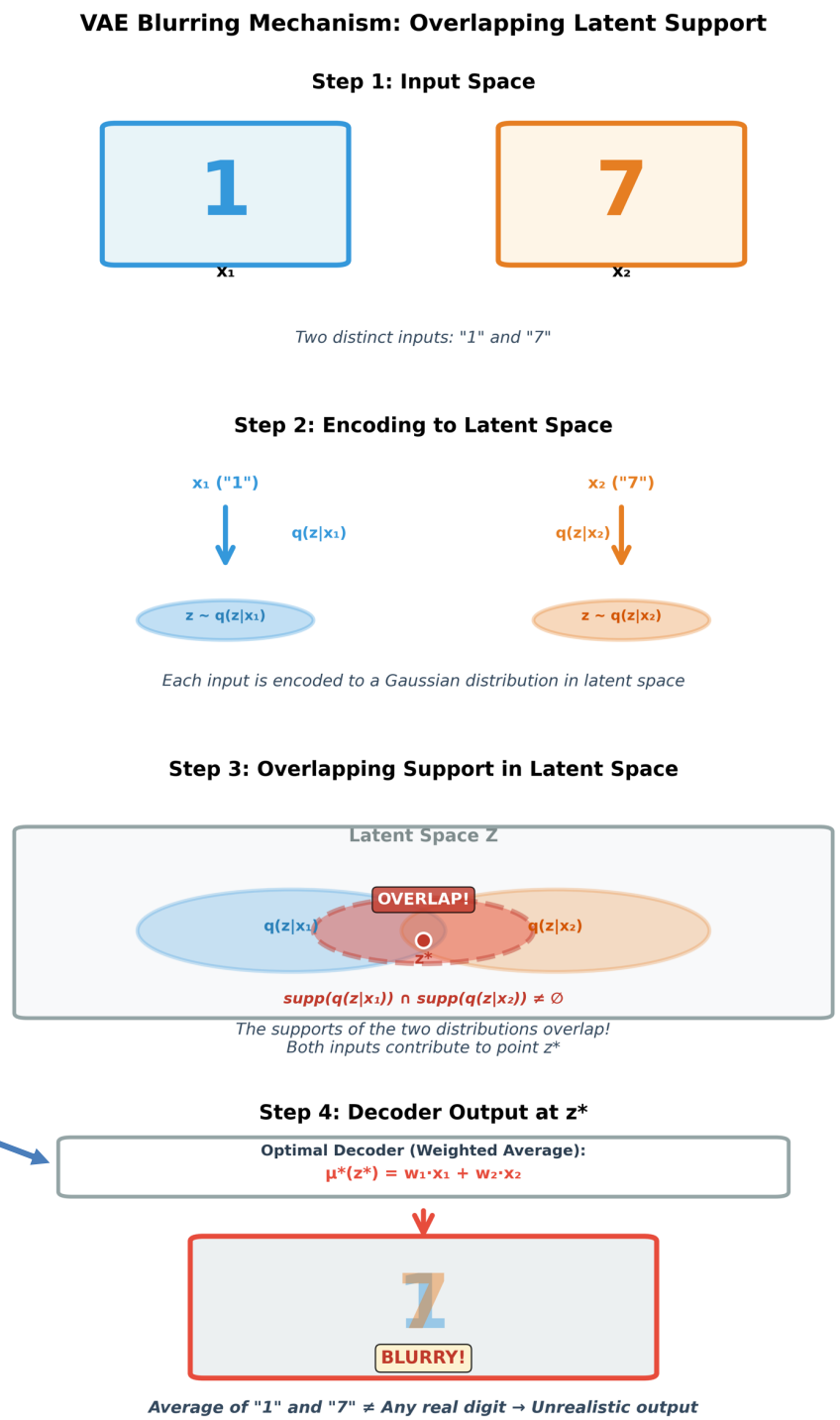
VAE's Blurry Generation Problem

Now suppose that two distinct inputs $\mathrm{x} \neq \mathrm{x}'$ are mapped to overlapping regions in latent space, i.e., the supports of $q_{enc}(\cdot \mid \mathrm{x})$ and $q_{enc}(\cdot \mid \mathrm{x}')$ intersect.

Then, $\mu^*(\mathrm{z})$ averages over multiple, potentially unrelated inputs, which leads to blurry, non-distinct outputs.

$$\boldsymbol{\mu^*(\mathrm{z}) = \frac{\mathbb{E}_{p_{data}(\mathrm{x})}[q_{\mathrm{enc}}(\mathbf{z} \mid \mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{data}(\mathrm{x})}[q_{\mathrm{enc}}(\mathbf{z} \mid \mathbf{x})]}}$$

This averaging effect over conflicting modes is a fundamental reason for the characteristic blurriness in VAE-generated samples.



Why Deeper Networks in a Flat VAE are Not Enough

Limit of Posterior Family

$$q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}\left(\mathbf{z}; \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}), \operatorname{diag}\left(\sigma_{\theta}^{2}(\mathbf{x})\right)\right)$$

- It is a single Gaussian with diagonal covariance
- Deeper network can improve the accuracy of $\boldsymbol{\mu}_{\boldsymbol{\theta}}$ and σ_{θ} but does not expand the family

$$\boxed{q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x}) \approx p_{\boldsymbol{\phi}}(\mathbf{z} \mid \mathbf{x})}$$

- When $p_{\boldsymbol{\phi}}(\mathbf{z} \mid \mathbf{x})$ is multi-peaked (or multi-modal), $q_{\boldsymbol{\theta}}(\mathbf{z} \mid \mathbf{x})$ cannot match it
- It loosens the ELBO and weakens inference.
- Addressing this requires a richer posterior class, not merely deeper networks

Why Deeper Networks in a Flat VAE are Not Enough

Posterior Collapse

- if the decoder is too expressive, the model may suffer from posterior collapse
- In order to explain posterior collapse, we should consider all samples ~ p_{data}

$$\begin{aligned}&\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\mathcal{L}_{\text{ELBO}}(\mathbf{x})] \\ &\quad = \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}[\log p_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\mathcal{D}_{\text{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))] \\ &\quad = \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}[\log p_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z})] - \mathcal{I}_q(\mathbf{x};\mathbf{z}) - \mathcal{D}_{\text{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z})\|p(\mathbf{z})),\end{aligned}$$

[posterior collapse]
- 모든 입력 x 가 동일한 z 분포로 매핑되는 현상
- $q_\theta(\mathbf{z}|\mathbf{x}) \approx q_\theta(\mathbf{z}) \approx p(\mathbf{z}) \quad \text{for all } \mathbf{x}$

where $I_q(\mathbf{x};\mathbf{z})$ is the mutual information defined by

$$I_q(\mathbf{x};\mathbf{z}) = \mathbb{E}_{q(\mathbf{x},\mathbf{z})}\left[\log \frac{q_{\boldsymbol{\theta}}(\mathbf{z}\mid\mathbf{x})}{q(\mathbf{z})}\right] = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\mathcal{D}_{\text{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z}\mid\mathbf{x})\|q(\mathbf{z}))]$$

and the aggregated posterior $q(\mathbf{z})$ is $q_{\boldsymbol{\theta}}(\mathbf{z}) = \int p_{data}(\mathbf{x})q_{\boldsymbol{\theta}}(\mathbf{z}\mid\mathbf{x})\mathrm{d}\mathbf{x}$

- a maximizer of the ELBO tries set $I_q(\mathbf{x};\mathbf{z}) = 0$ and $q_{\boldsymbol{\theta}}(\mathbf{z}) = p(\mathbf{z})$.
- This "ignore z (posterior collapse)" solution does not disappear by making the networks deeper

From Standard VAE to Hierarchical VAEs

Hierarchical Variational Autoencoders (HVAEs)

HVAEs Summary

- Purpose: Extend VAE hierarchically for complex data modeling
- Structure: Introduce multiple layers of latent variable hierarchy
- Advantage: Capture data features at multiple abstraction levels → significantly boost expressive power
- Characteristic: Naturally mirror the compositional nature of real-world data

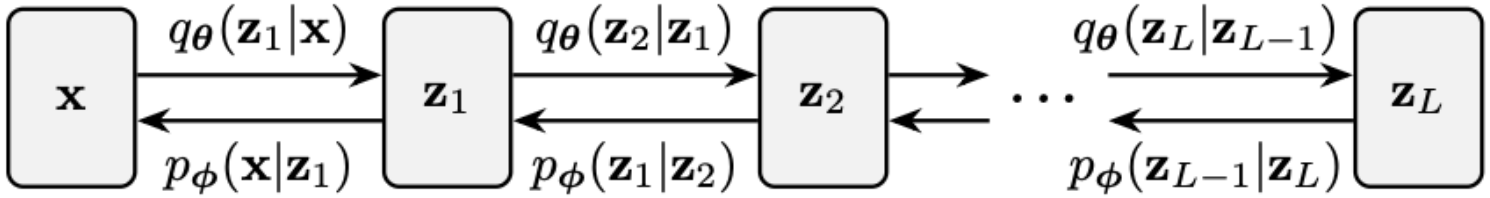


Figure 2.2: Computation graph of the HVAE. It has a hierarchical structure with stacked, trainable encoders and decoders across multiple latent layers.

HVAE's Modeling

HVAEs Decoder Structure

- Architecture: Multiple latent layers $p_\phi(z_{i-1} \mid z_i)$ in top-down hierarchy
- Chain: Each layer conditions the next, forming conditional priors
- Abstraction: Progressively finer levels (coarse → fine)

Bottom (Down) Top (Up)

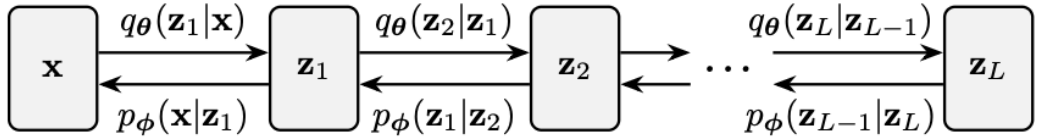


Figure 2.2: Computation graph of the HVAE. It has a hierarchical structure with stacked, trainable encoders and decoders across multiple latent layers.

Abstraction: Progressively finer levels (coarse → fine)

Sample $\mathbf{z}_L$ from prior, sequentially decoded down ($\mathbf{z}_L \to \mathbf{z}_{L-1} \to \cdots \to \mathbf{z}_1$), finally to the final observation $\mathbf{x}$

This leads to the following top-down factorization of the joint distribution

$$p_\phi(\mathbf{x}, \mathbf{z}_{1:L}) = p_\phi(\mathbf{x} \mid \mathbf{z}_1) \prod_{i=2}^{L} p_\phi(\mathbf{z}_{i-1} \mid \mathbf{z}_i) p(\mathbf{z}_L).$$

This structure defines the marginal data distribution

$$p_{\text{HVAE}}(\mathbf{x}) := \int p_\phi(\mathbf{x}, \mathbf{z}_{1:L}) \mathrm{d}\mathbf{z}_{1:L}$$

HVAE Encoder Structure

Learnable encoder $q_\theta(\mathbf{z}_{1:L} \mid \mathbf{x})$ mirrors generative hierarchy, typically using bottom-up Markov factorization

$$q_\theta(\mathbf{z}_{1:L} \mid \mathbf{x}) = q_\theta(\mathbf{z}_1 \mid \mathbf{x}) \prod_{i=2}^{L} q_\theta(\mathbf{z}_i \mid \mathbf{z}_{i-1})$$

HAVE's Characteristics

HVAE ELBO Derivation

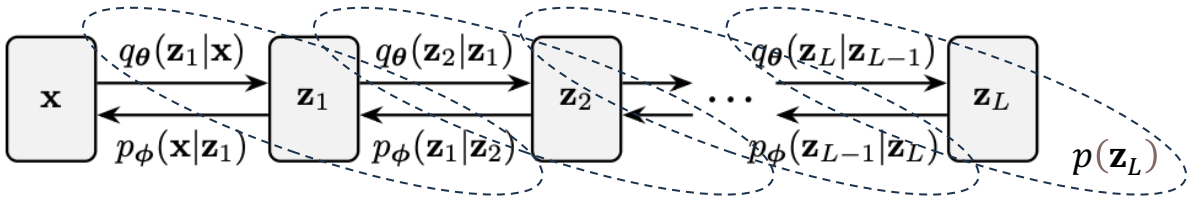
$$
\begin{aligned}
\log p_{\mathrm{HVAE}}(\mathbf{x}) &= \log \int p_{\boldsymbol{\phi}}(\mathbf{x}, \mathbf{z}_{1:L})\, \mathrm{d}\mathbf{z}_{1:L} \\
&= \log \int \frac{p_{\boldsymbol{\phi}}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})\, \mathrm{d}\mathbf{z}_{1:L} \\
&= \log \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} \left[\frac{p_{\boldsymbol{\phi}}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} \right] \\
&\geq \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} \left[\log \frac{p_{\boldsymbol{\phi}}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} \right] \\
&=: \mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{\phi}).
\end{aligned}
\tag{2.1.3}
$$

$$
\mathcal{L}_{\mathrm{ELBO}} = \mathbb{E}_{q_{\boldsymbol{\theta}}(\mathbf{z}_{1:L}|\mathbf{x})} \left[\log \frac{p(\mathbf{z}_L) \prod_{i=2}^{L} p_{\boldsymbol{\phi}}(\mathbf{z}_{i-1}|\mathbf{z}_i) p_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z}_1)}{q_{\boldsymbol{\theta}}(\mathbf{z}_1|\mathbf{x}) \prod_{i=2}^{L} q_{\boldsymbol{\theta}}(\mathbf{z}_i|\mathbf{z}_{i-1})} \right]
$$

$$
\begin{aligned}
&= \mathbb{E}_q[\log p_{\boldsymbol{\phi}}(\mathbf{x}|\mathbf{z}_1)] - \mathbb{E}_q \left[\mathcal{D}_{\mathrm{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z}_1|\mathbf{x}) \| p_{\boldsymbol{\phi}}(\mathbf{z}_1|\mathbf{z}_2)) \right] \\
&\quad - \sum_{i=2}^{L-1} \mathbb{E}_q \left[\mathcal{D}_{\mathrm{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z}_i|\mathbf{z}_{i-1}) \| p_{\boldsymbol{\phi}}(\mathbf{z}_i|\mathbf{z}_{i+1})) \right] \\
&\quad - \mathbb{E}_q \left[\mathcal{D}_{\mathrm{KL}}(q_{\boldsymbol{\theta}}(\mathbf{z}_L|\mathbf{z}_{L-1}) \| p(\mathbf{z}_L)) \right].
\end{aligned}
$$

- This spreads the information penalty across multiple levels and localizes learning signals via adjacent KL terms.
- These benefits arise from the hierarchical latent structure itself, not merely from making a flat VAE deeper.

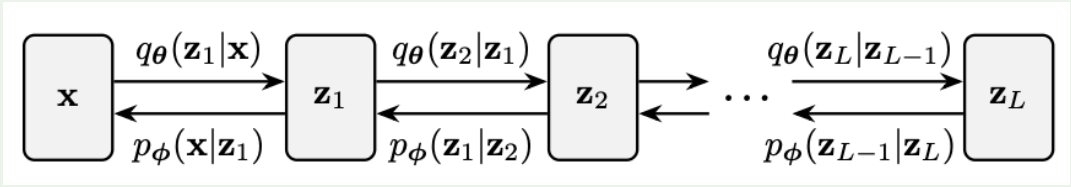
Each inference conditional is aligned with its top-down generative counterpart



HAVE's Impact & Challenge

HVAE Impact

- Stacked layers enable progressive generation from coarse to fine details.



- This hierarchy simplifies modeling complex high-dimensional data.
- It is a key principle behind modern generative models like score-based methods and normalizing flows.

HAVE Challenge

- Lower layers dominate reconstruction, leaving higher latents under-utilized.
- Gradients to deep latents are indirectly learned and weak.

So, Diffusion Model...

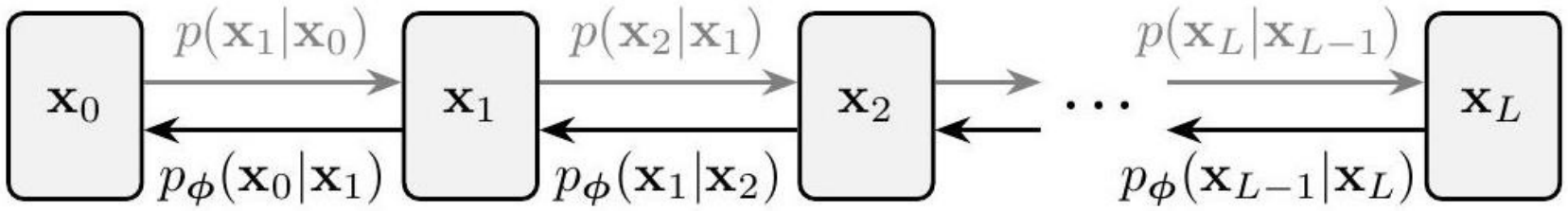
- So, diffusion models retain hierarchy while avoiding these issues.
- Fixing encoding and learning only the reverse process improves stability and generation quality

Variational Perspective: Denoising Diffusion Probabilistic Models (DDPM)

DDPM Overview and Core Concepts

Denoising Diffusion Probabilistic Models (DDPMs)

It represent a cornerstone of diffusion modeling.
They operate within a variational framework, much like VAEs and HVAEs, but with a clever twist.



→ Forward Pass

Fixed Encoder: Gradually corrupts data by injecting G aussian noise

Fixed transition kernel: $p(\mathbf{x}_i \mid \mathbf{x}_{i-1})$

Data evolves into an isotropic (모든 차원의 분산이 동일) Gaussian distribution (i.e., $\mathcal{N}(\mu, \sigma^2\mathbf{I})$)

← Reverse Denoising Process

Learnable Decoder: Neural network learns to r everse noise corruption

Parameterized distribution: $p_\phi(\mathbf{x}_{i-1} \mid \mathbf{x}_i)$

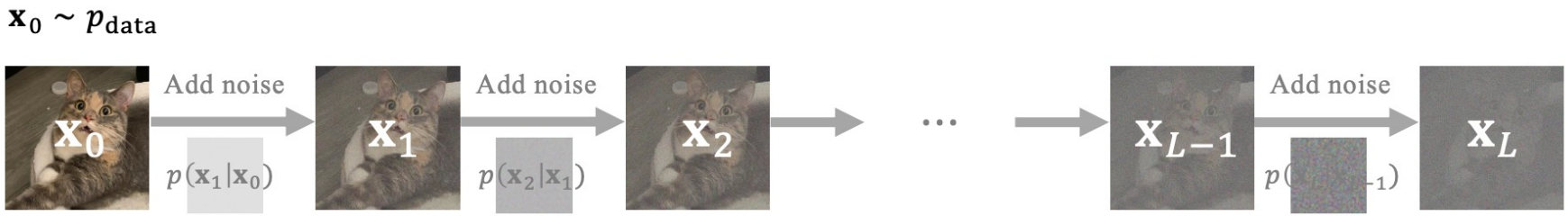
Each denoising step is a manageable task

DDPM – Forward Process (Fixed Encoder)

Forward Process – Fixed Encoder

The forward process is a fixed, non-trainable operation that serves as an encoder

It progressively corrupts data by adding noise, transforming it into a simple prior distribution $p_{prior} := \mathcal{N}(0, \boldsymbol{I})$



Fixed Gaussian Transitions

$$p(\mathbf{x}_i \mid \mathbf{x}_{i-1}) := \mathcal{N}\left(\mathbf{x}_i; \sqrt{1-\beta_i^2}\mathbf{x}_{i-1}, \beta_i^2 \mathbf{I}\right)$$

- Process begins with $\mathbf{x}_0$, representing a sample drawn from the real data distribution p_{data}
- Sequence $\{\beta_i\}_{i=1}^{L}$ denotes a pre-determined, monotonically increasing noise schedule, where each $\beta_i \in (0,1)$ controls the variance injected at step i

Iterative Update $\alpha_i := \sqrt{1-\beta_i^2}$

$$\mathbf{x}_i = \alpha_i \mathbf{x}_{i-1} + \beta_i \boldsymbol{\epsilon}_i \quad \text{where } \boldsymbol{\epsilon}_i \sim \mathcal{N}(0, I)$$

- Reparameterization trick
- At each step i, we scale down the previous state $\mathbf{x}_{i-1}$ by α_i and add a controlled amount of Gaussian noise scaled by β_i

Forward Process – Perturbation Kernel and Prior Distribution

Direct Sampling at Step i

- a closed-form expression for the distribution of noisy samples at step i given the original data $\mathbf{x}_0$

$$p_i(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_i; \bar{\alpha}_i \mathbf{x}_0, \left(1 - \bar{\alpha}_i^2\right)\mathbf{I}\right)$$

where

$$\bar{\alpha}_i \coloneqq \prod_{k=1}^{i} \sqrt{1 - \beta_k^2} = \prod_{k=1}^{i} \alpha_k.$$

- This means we can sample $\mathbf{x}_i$ directly from $\mathbf{x}_0$ as

$$\mathbf{x}_i = \bar{\alpha}_i \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_i^2}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$$

Converge to Prior distribution

- Let the noise schedule $\{\beta_i\}_{i=1}^{L}$ be an increasing sequence, then the marginal distribution of the forward process converges as

$$p_L(\mathbf{x}_L \mid \mathbf{x}_0) \longrightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}) \text{ as } L \to \infty$$

- It motivates the choice of the prior distribution as

$$p_{prior} \coloneqq \mathcal{N}(0, \boldsymbol{I})$$

Forward Process – Perturbation Kernel and Prior Distribution

1 한 줄 정의부터

α_i와 β_i는
forward diffusion 과정에서 "신호(signal)"와 "노이즈(noise)"의 비율을 정하는 스케줄 파라미터야.

- α_i : "이전 그림을 얼마나 덜 흐리게 남길지"
- β_i : "이번 단계에서 물감을 얼마나 더 흩뿌릴지"

뒤로 갈수록
- $\alpha_i \downarrow$
- $\beta_i \uparrow$

그래서 점점 완전한 노이즈가 됨.

2 Forward diffusion에서의 정의

DDPM의 forward process는 이렇게 정의돼:

$$x_i = \alpha_i x_{i-1} + \beta_i \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, I)$$

즉,

- α_i : **이전 상태를 얼마나 보존할지**
- β_i : **새 노이즈를 얼마나 추가할지**

를 결정하는 계수야.

3 왜 $\alpha_i^2 + \beta_i^2 = 1$ 인가?

보통 DDPM에서는

$$\alpha_i^2 = 1 - \beta_i^2$$

로 정의함. 이유는:

- 분산이 폭주하지 않게 하려고
- 각 step이 "정규화된 가우시안 노이즈 추가"가 되도록

즉, 에너지(분산)를 안정적으로 유지하기 위한 설계야.

4 $\bar{\alpha}_i$의 의미 (누적 신호량)

$$\bar{\alpha}_i = \prod_{k=1}^{i} \alpha_k$$

👉 x_0 (clean data)가
x_i에 **얼마나 남아 있는지**를 나타냄.

그래서 forward marginal은:

$$x_i = \bar{\alpha}_i x + \sqrt{1 - \bar{\alpha}_i^2}\, \varepsilon$$

이 형태가 나옴.

$p_i(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_i; \bar{\alpha}_i \mathbf{x}_0, \left(1 - \bar{\alpha}_i^2\right)\mathbf{I}\right)$ → 유도 과정 상세 설명

1. 기본 설정

재귀 관계식:

$$\mathbf{x}_i = \alpha_i \mathbf{x}_{i-1} + \beta_i \boldsymbol{\epsilon}_i, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

여기서 $\alpha_i = \sqrt{1 - \beta_i^2}$ 이고, 모든 $\boldsymbol{\epsilon}_i$ 는 **독립적**입니다.

2. 단계별 전개

Step 1: $\mathbf{x}_1$ 표현

$$\mathbf{x}_1 = \alpha_1 \mathbf{x}_0 + \beta_1 \boldsymbol{\epsilon}_1$$

분포로 쓰면:

$$\mathbf{x}_1 | \mathbf{x}_0 \sim \mathcal{N}(\alpha_1 \mathbf{x}_0, \beta_1^2 \mathbf{I})$$

Step 2: $\mathbf{x}_2$ 표현

$$\mathbf{x}_2 = \alpha_2 \mathbf{x}_1 + \beta_2 \boldsymbol{\epsilon}_2$$

$$\mathbf{x}_2 = \alpha_2(\alpha_1 \mathbf{x}_0 + \beta_1 \boldsymbol{\epsilon}_1) + \beta_2 \boldsymbol{\epsilon}_2$$

$$\mathbf{x}_2 = \alpha_2 \alpha_1 \mathbf{x}_0 + \alpha_2 \beta_1 \boldsymbol{\epsilon}_1 + \beta_2 \boldsymbol{\epsilon}_2$$

이제 **가우시안 노이즈 항들을 하나로 합쳐야** 합니다.

핵심: 독립 가우시안의 선형 결합

$\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2$ 가 독립이므로:

- $\alpha_2 \beta_1 \boldsymbol{\epsilon}_1 \sim \mathcal{N}(\mathbf{0}, (\alpha_2 \beta_1)^2 \mathbf{I})$
- $\beta_2 \boldsymbol{\epsilon}_2 \sim \mathcal{N}(\mathbf{0}, \beta_2^2 \mathbf{I})$

독립적인 가우시안 변수의 합:

$$\alpha_2 \beta_1 \boldsymbol{\epsilon}_1 + \beta_2 \boldsymbol{\epsilon}_2 \sim \mathcal{N}(\mathbf{0}, [(\alpha_2 \beta_1)^2 + \beta_2^2]\mathbf{I})$$

분산을 계산하면:

$$\text{Var} = (\alpha_2 \beta_1)^2 + \beta_2^2 = \alpha_2^2 \beta_1^2 + \beta_2^2$$

3. 분산 간소화의 핵심 트릭

$\alpha_i = \sqrt{1 - \beta_i^2}$ 라는 정의를 사용하면:

$$\alpha_i^2 = 1 - \beta_i^2 \quad \Rightarrow \quad \beta_i^2 = 1 - \alpha_i^2$$

따라서:

$$\text{Var} = \alpha_2^2 \beta_1^2 + \beta_2^2$$

$$= \alpha_2^2 (1 - \alpha_1^2) + (1 - \alpha_2^2)$$

$$= \alpha_2^2 - \alpha_2^2 \alpha_1^2 + 1 - \alpha_2^2$$

$$= 1 - \alpha_2^2 \alpha_1^2$$

$$= 1 - (\alpha_1 \alpha_2)^2$$

결과:

$$\mathbf{x}_2 = \alpha_1 \alpha_2 \mathbf{x}_0 + \sqrt{1 - (\alpha_1 \alpha_2)^2}\, \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

따라서:

$$\mathbf{x}_2 | \mathbf{x}_0 \sim \mathcal{N}((\alpha_1 \alpha_2)\mathbf{x}_0, [1 - (\alpha_1 \alpha_2)^2]\mathbf{I})$$

$p_i(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_i; \bar{\alpha}_i \mathbf{x}_0, \left(1 - \bar{\alpha}_i^2\right)\mathbf{I}\right)$ → 유도 과정 상세 설명

Step 3: $\mathbf{x}_3$ 표현

같은 방식으로:

$$\mathbf{x}_3 = \alpha_3 \mathbf{x}_2 + \beta_3 \boldsymbol{\epsilon}_3$$

$$= \alpha_3(\alpha_1\alpha_2\mathbf{x}_0 + \sqrt{1-(\alpha_1\alpha_2)^2}\boldsymbol{\epsilon}) + \beta_3\boldsymbol{\epsilon}_3$$

$$= \alpha_1\alpha_2\alpha_3\mathbf{x}_0 + \alpha_3\sqrt{1-(\alpha_1\alpha_2)^2}\boldsymbol{\epsilon} + \beta_3\boldsymbol{\epsilon}_3$$

노이즈 항들의 분산:

$$\text{Var} = \alpha_3^2[1-(\alpha_1\alpha_2)^2] + \beta_3^2$$

$$= \alpha_3^2[1-(\alpha_1\alpha_2)^2] + (1-\alpha_3^2)$$

$$= \alpha_3^2 - \alpha_3^2(\alpha_1\alpha_2)^2 + 1 - \alpha_3^2$$

$$= 1 - (\alpha_1\alpha_2\alpha_3)^2$$

Direct Sampling at Step i

- a closed-form expression for the distribution of noisy samples at step i given the original data $\mathbf{x}_0$

$$p_i(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_i; \bar{\alpha}_i \mathbf{x}_0, \left(1 - \bar{\alpha}_i^2\right)\mathbf{I}\right)$$

where

$$\bar{\alpha}_i := \prod_{k=1}^{i} \sqrt{1-\beta_k^2} = \prod_{k=1}^{i} \alpha_k.$$

- This means we can sample $\mathbf{x}_i$ directly from $\mathbf{x}_0$ as

$$\mathbf{x}_i = \bar{\alpha}_i \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_i^2}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I) \qquad \text{[2.2.1]}$$

$p_i(\mathbf{x}_i \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_i; \bar{\alpha}_i \mathbf{x}_0, \left(1 - \bar{\alpha}_i^2\right)\mathbf{I}\right)$ → 유도 과정 상세 설명

4. 직접 샘플링 공식 (2.2.1)

위 결과로부터 **임의의 timestep i에서 직접 샘플링** 가능:

$$\mathbf{x}_i = \bar{\alpha}_i \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_i^2}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

이것이 중요한 이유:

- 중간 스텝들을 거치지 않고 바로 $\mathbf{x}_0$에서 $\mathbf{x}_i$를 생성 가능!
- 학습 시 효율성 크게 향상

구체적 예시:

β 스케줄이 다음과 같다고 가정:

- $\beta_1 = 0.1, \beta_2 = 0.2, \beta_3 = 0.3$

그러면:

- $\alpha_1 = \sqrt{1 - 0.01} = 0.995$
- $\alpha_2 = \sqrt{1 - 0.04} = 0.980$
- $\alpha_3 = \sqrt{1 - 0.09} = 0.954$

$\mathbf{x}_3$을 직접 샘플링하려면:

$$\bar{\alpha}_3 = 0.995 \times 0.980 \times 0.954 \approx 0.930$$

$$\mathbf{x}_3 = 0.930 \cdot \mathbf{x}_0 + \sqrt{1 - 0.930^2}\boldsymbol{\epsilon} = 0.930 \cdot \mathbf{x}_0 + 0.367\boldsymbol{\epsilon}$$

해석: 원본의 93%를 유지하고, 표준편차 0.367의 노이즈를 추가

5. Prior Distribution으로의 수렴

노이즈 스케줄 $\{\beta_i\}_{i=1}^{L}$이 증가하는 시퀀스이면:

$$p_L(\mathbf{x}_L|\mathbf{x}_0) \longrightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \text{as } L \to \infty$$

왜냐하면:

- L이 커질수록 $\bar{\alpha}_L = \prod_{k=1}^{L} \alpha_k \to 0$
- 따라서 $\mathbf{x}_L \approx \sqrt{1 - 0^2}\boldsymbol{\epsilon} = \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

이것이 **prior distribution**을 $p_{\text{prior}} := \mathcal{N}(\mathbf{0}, \mathbf{I})$로 선택하는 이유입니다!